

Q1: 15 points)

$$1a) : X[k] = \sum_{n=0}^3 x[n] W_4^{nk} = 1 - (-1)^k$$

$$1b) : H[k] = \sum_{n=0}^3 h[n] W_4^{nk} = \frac{15}{2 \cdot j^{-k} - 1}$$

$$1c) : y[n] = -3\delta[n] - 6\delta[n-1] + 3\delta[n-2] + 6\delta[n-3]$$

$$1d) : y[n] = -3\delta[n] - 6\delta[n-1] + 3\delta[n-2] + 6\delta[n-3]$$

Q2: 15 points)

1a) : 1° : When $(m-k) \mid (N+1)$, $\exists r \in \mathbb{Z}$, s.t. $m-k = r(N+1)$

$$\begin{aligned} \text{LHS} &= \sum_{n=-N/2}^{N/2} W_{N+1}^{-n(m-k)} \\ &= \sum_{n=-N/2}^{N/2} e^{j2\pi \frac{n(m-k)}{N+1}} \\ &= \sum_{n=-N/2}^{N/2} e^{j2\pi nr} \\ &= \sum_{n=-N/2}^{N/2} 1 \\ &= N+1 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= (N+1) \sum_{r=-\infty}^{+\infty} \delta[m-k-r(N+1)] \\ &= N+1 \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS} = 1$$

2° : When $(m-k) \nmid (N+1)$, $\exists r \notin \mathbb{Z}$, s.t. $m-k = r(N+1)$

$$\begin{aligned} \text{LHS} &= \sum_{n=-N/2}^{N/2} W_{N+1}^{-n(m-k)} \\ &= \sum_{n=-N/2}^{N/2} e^{j2\pi \frac{n(m-k)}{N+1}} \end{aligned}$$

$$\begin{aligned} \text{RHS} &= (N+1) \sum_{r=-\infty}^{+\infty} \delta[m-k-r(N+1)] \\ &= 0 \end{aligned}$$

$$\begin{aligned} &= \sum_{n=-N/2}^{N/2} e^{j2\pi nr} \\ &= \frac{e^{j2\pi r(-N/2)} - e^{j2\pi r(N/2+1)}}{1 - e^{j2\pi r}} \\ &= 0 \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS} = 0$$

Q. E. D.

$$(b): F_k = \frac{1}{N+1} \sum_{n=-N/2}^{N/2} f_n W_{N+1}^{-nk}$$

$$\text{let } f_n = C \cdot \sum_{k=-N/2}^{N/2} F_k W_{N+1}^{nk}$$

From the results of part (a), we have:

$$f_n = C \cdot \frac{1}{N+1} \sum_{k=-N/2}^{N/2} \left(\sum_{m=-N/2}^{N/2} f_m W_{N+1}^{-mk} \right) W_{N+1}^{nk}$$

$$= \frac{C}{N+1} \sum_{m=-N/2}^{N/2} \left(f_m \sum_{k=-N/2}^{N/2} W_{N+1}^{-k(m-n)} \right)$$

$$= \frac{C}{N+1} \sum_{m=-N/2}^{N/2} \left(f_m (N+1) \sum_{r=-\infty}^{+\infty} \delta[m-n-r(N+1)] \right)$$

$$= C \cdot \sum_{m=-N/2}^{N/2} \left(f_m \sum_{r=-\infty}^{+\infty} \delta[m-n-r(N+1)] \right)$$

$$= C \cdot f_n \quad (\text{only when } m=n \text{ and } r=0, \delta \text{ function is nonzero})$$

$$\Rightarrow C=1$$

$$\therefore f_n = \sum_{k=-N/2}^{N/2} F_k W_{N+1}^{nk}$$

(c): Index of x_n ranges from $-N/2$ to $N/2$

$$X_{-N/2} = -\frac{NA}{2(N+1)}, \quad X_{N/2} = \frac{NA}{2(N+1)}$$

$$\lim_{N \rightarrow \infty} X_{-N/2} = -\frac{A}{2}, \quad \lim_{N \rightarrow \infty} X_{N/2} = \frac{A}{2}$$

Q3: (5 points):

$$\begin{cases} \tilde{x}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] W_N^{-kn} \\ \tilde{X}[k] = \sum_{n=0}^{N-1} \tilde{x}[n] W_N^{kn} \end{cases}$$

From the sequence $\tilde{f}_n$, we have

$$\tilde{F}_k = \{-1 + (-2 + 2\sqrt{2})j, 1, -1 + (2 + 2\sqrt{2})j, 1, -1 + (-2 - 2\sqrt{2})j, 1, -1 + (2 - 2\sqrt{2})j\}$$

$k = -3:4$, and other points are all zero. obviously it's not odd and imaginary

The reason is that for $t=0$ and $\pm \frac{1}{2}$, they are breaking points.
To correct the definition, the sampling value

To correct the input sequence, let sampling point $n' = -3:1:3$ ($N=6$)

$$f_n' = \{-1, -1, -1, 0, 1, 1, 1\}$$

$$F_k = \sum_{n=-N/2}^{N/2} f_n W_{N+1}^{nk}$$

$$F_{-k} = \sum_{n=-N/2}^{N/2} f_n W_{N+1}^{-nk}$$

$$= \sum_{n=-N/2}^{N/2} f_{-n} W_{N+1}^{-nk}$$

$$= -F_k$$

in this way F_k is odd and imaginary

Q4: (5 points)

1a): Assume ~~for~~ $x[n] = 1$ for $n = 0:49$; $h[n] = 1$ for $n = 0:9$

To avoid aliasing, $x_{zp} = \begin{cases} x[n], & n = 0:49 \\ 0, & n = 50:58 \end{cases}$ $h_{zp} = \begin{cases} h[n], & n = 0:9 \\ 0, & n = 10:58 \end{cases}$

$$\Rightarrow y[n] = x[n] * h[n] = \begin{cases} x[n] \otimes h[n], & n = 0:58 \\ 0, & \text{o.w.} \end{cases}$$

$\therefore$ The number of nonzero points is 59.

1b): $y_c[n] = x[n] \otimes h[n] = 10, n = 0:49$

The nonzero part of $y[n]$ has aliasing at $y_c[n]$

$$\therefore y_c[n] = \begin{cases} y[n] + y[n+50], & n = 0:8 \\ y[n], & n = 9:49 \end{cases}$$

and $y[n] = x[n] * h[n] = 5, n = 0:4$ $\Rightarrow y[n] = \begin{cases} 5 & n = 0:4 \\ 10 & n = 9:49 \\ 5 & n = 50:54 \end{cases}$

and $y[5] + y[55] = y[6] + y[56] = y[7] + y[57] = y[8] + y[58] = 10$

Among $n = 0:58$, we have 51 determined points, 8 points undetermined.